LOVELY TECH POS

Requirements and Workflow Baseline

Reconciled edition | v2.1 | 9 October 2026

Normative scope

This edition includes the complete canonical requirements, preserves valid workflow details from the v2.0 source PDF, and adds release acceptance gates. Where the old PDF conflicts with the canonical requirements, the canonical requirements prevail.

Session policy reconciled

- 30-minute idle timeout
- 12-hour absolute session lifetime
- 10-minute critical reauthentication window

Source documents

docs/REQUIREMENTS.md

docs/RELEASE_CHECKLIST.md

Requirements define intended behavior; acceptance evidence is not implied.

This document records the agreed product scope. It is normative: implementation status is tracked separately in PROJECT_STATUS.md, and release evidence is tracked in RELEASE_CHECKLIST.md.

1. Architecture and technical boundaries

- Use PostgreSQL 18, Python 3.13, FastAPI, SQLAlchemy 2.x, Alembic, React, TypeScript, Vite, and Tauri 2. Communicate through REST/JSON over the local network.
- All database access is server-side through FastAPI. Client applications must never connect directly to PostgreSQL.
- Use UUIDs for internal identities and backend-allocated human-readable document numbers. Number allocation must be atomic and invoice numbers must never be reused.
- Represent money with Python Decimal and PostgreSQL fixed-point DECIMAL/NUMERIC, never binary floating point.
- Inventory is movement- and lot-based. Cost of goods sold follows exact FIFO, including deterministic handling and reconciliation of negative stock.
- Enforce permissions and data visibility on the backend. Hiding controls in the UI is not an authorization boundary.
- Preserve audit and business-history records as append-only. Important transactions must not be hard-deleted.
- Use idempotency keys for retryable financial mutations, concurrency/version checks for controlled edits, and branch isolation for business data.
- Integrate local printers without making transaction success depend on printer availability.
- Support encrypted local backups and optional Google Drive backup storage. Google Drive availability must not be required for ordinary LAN POS operation.
- Use Pytest for backend regression tests and Playwright for browser acceptance tests.
- The Cashier laptop is the always-on production server (PostgreSQL plus FastAPI). The Owner/Admin laptop is a client and may be turned off without interrupting POS operations.
- The same supported installer/executable should offer server or client roles. Clients must discover the current server without a hardcoded IP. Replacement-server recovery must restore the latest VERIFIED backup with minimal manual configuration.
- Normal installation and operation must not depend on Git.

2. Security, sessions, and role permissions

- Cashiers must never access purchase/supplier cost, FIFO unit cost, profit, internal pricing rationale, or detailed backup/restore information. Enforce this in API permissions, response schemas, and queries as appropriate—not only by hiding UI.
- Required session controls: 30-minute idle timeout, 12-hour absolute lifetime, and a 10-minute critical reauthentication window. Reauthentication must verify the relevant user's password and be scoped to authorized critical actions.
- Preserve existing RBAC, branch isolation, Admin recovery controls, and permission revocation behavior.
- Record security-sensitive and financial actions in the append-only audit/history system, including Admin approvals, overrides, corrections, reprints, backup/restore, and recovery.
- Apply stable idempotency keys to retryable create/payment/refund/settlement actions. Replaying an identical request must not create duplicate financial movements; conflicting payloads under the same key must be rejected.
- Protect mutable records with optimistic version checks and transactional locking where concurrent updates could violate business or financial invariants.
- Do not claim full security, operating-system hardening, disk encryption, or recovery guarantees without direct evidence and acceptance testing.

3. Sales, payments, and documents

- Support Cash, Card, Online, and split payments.
- Permit typed sale prices, with configurable cashier discount limits and Admin approval/override for amounts beyond authority.
- Credit sales are Admin-only.
- Assign invoice numbers atomically and never reuse them.
- Commit the sale before printing. Printer failure must not roll back a committed sale.
- Reprinting is a separate document event with an audit trail.
- Duplicate clicks, request timeouts, and safe retries must not create duplicate sales or payments.
- Preserve sale-time price, discount, payment, customer, invoice, and warranty snapshots needed for historical documents and reports.

4. Inventory and FIFO

- Track inventory through movements and purchase lots; calculate cost using exact FIFO.
- Negative stock may be allowed only with a clear warning and deterministic later reconciliation. A later purchase retains its full original quantity and cost.
- Reconcile negative allocations deterministically without changing historical purchase-lot identity or quantity.
- Audit purchase cost, date, quantity, and reference corrections, and recalculate affected FIFO allocations and dependent costs correctly.
- Preserve original stock lineage for valid returns and reversals. Damage/loss must be represented explicitly rather than silently returned to saleable stock.

5. Returns, exchanges, and purchases

- The normal return window is 7 days. Beyond-authority returns require an authorized Admin override. Support partial returns.
- GOOD returns restore the original stock lineage. Damaged returns route to Damage/Loss.
- Return charges are separate income/profit entries. An exchange is a separate return plus a separate new sale; refund and payment accounting must remain correct.
- Cashiers may enter purchase items and quantities without seeing cost. Admins have a private pending-cost workflow; completed costs create purchase lots.
- Corrections to completed purchases are audited and must obey concurrency checks and FIFO lineage safety.

6. Repairs

- Provide a repair intake form, labour-only charges, FIFO part consumption, and original-lineage restoration when an unused GOOD part is returned.
- Route damaged parts to Damage Review/Admin.
- Support advances, final payments, outstanding balances, Cash/Card/Online/split payment, and an Admin delivery override.
- Cancellation reverses stock effects safely while preserving immutable payment history.
- Preserve repair references, estimates/final charges, customer and device details, status history, and printable/reprintable records.

7. Warranty

- Warranty begins on the sale date and stores exact expiry/status plus any authorized sale-time override.

- Support NONE, DAYS, MONTHS, CHECKING ONLY, and CUSTOM rules.
- Warranty claims are separate from repairs. Cashiers may create claims; Admins decide supplier, damage, replacement, and other permitted outcomes.
- Supplier replacement is recorded as WARRANTY_REPLACEMENT, not as a purchase.
- Support rejection paths, exception-only manual registration, lost-bill recovery, and audited reprints.
- Preserve immutable sale-time warranty snapshots so later product-default changes do not rewrite a completed sale's warranty.

8. Cash and business day

- Opening denominations are 5000, 2000, 1000, 500, 100, 50, and 20.
- Cash Drawer = opening + drawer inflows - refunds/outflows + Day-End reload.
- Cash In Hands = all approved shop inflows - all shop outflows, including every payment method.
- Expenses and purchases affect Cash In Hands but do not automatically reduce the physical Day-End drawer.
- Shift End is distinct from Day End.
- After Cashier Shift End, Admin has approximately one hour to Continue Today; otherwise the business day auto-closes. Admin reopening is permission-controlled and audited.
- Reload is one entry per day, brand-wise, cash-only, and Day-End related.
- Day End requires denomination reconciliation and a reason for any difference.

9. Expenses, other income, customers, and settlements

- Cashier-entered expenses and other income remain Pending Approval and are excluded from final Profit/Cash In Hands until approved.
- Admin approve/reject actions are audited; Admin may add categories.
- Customer details are optional where permitted. Customer history should include sales, repairs, warranty claims, bills, and settlements, subject to role-specific privacy.
- Credit sales are Admin-only.
- Repair balance settlements are payment events, not new revenue. Preserve settlement proof and reprint history.

10. Reports and targets

- Provide Daily, Monthly, Yearly, and Custom reporting for Sales, Profit, Stock, Purchase, Repair, Warranty, Cash Drawer, and Cash In Hands.
- Include payment splits, previous-month comparison, and PDF/Excel export where supported.
- Support targets for Sales Amount, Item Sales Profit, Repair Income, Repair Profit, Bills Count, and Items Sold Count.
- Cashiers must not see numeric profit targets. Progress may exceed 100%; visual bars may cap at 100%. Retain target history as immutable period snapshots.

11. Backup and disaster recovery

- Track the backup lifecycle as CREATED -> ENCRYPTED -> UPLOADED -> VERIFIED, with explicit failure/pending states where necessary.
- Only VERIFIED backups may be restored.
- Restore sequence: Admin reauthentication -> verified safety backup -> explicit confirmation -> selected verified restore point -> integrity check -> audit event.
- Queue and retry backup uploads after connectivity returns; recovery design must avoid split brain.
- Store emergency local backups separately from the database working directory.

- Cashier Shift End and Day End require a successfully verified backup. Retry a failed network backup; if the network target remains unavailable, use a separately configured local, USB, external, or network fallback target and verify the backup there before allowing close. Queue any deferred Google Drive upload for retry when connectivity returns.
- Google Drive is optional backup/recovery storage; normal LAN POS operation must remain available without cloud access.
- Protect and document the recovery key/secret. Losing or changing the key must not be represented as harmless if it makes old backups undecryptable.

12. Historical data import

- Support controlled import of products, purchase history, sales history, and customer/history from spreadsheets.
- Required workflow: staging -> validation -> mapping -> duplicate detection -> business-rule validation -> Admin preview -> approved import -> PostgreSQL write -> verification/reconciliation.
- Preserve original dates, references/invoices, quantities, costs/prices, customer/payment data where available, and provenance.
- Never fabricate missing historical records; unresolved data must be reported for human disposition.
- Reconcile opening stock + historical purchases - historical sales + returns/adjustments against current stock.
- Complete and approve historical import before final release/real-shop acceptance.

13. Accessibility, resilience, and release quality

- Provide clear validation, confirmation, timeout, connectivity, and recovery errors without disclosing credentials, tokens, or secrets.
- Protect irreversible and financially significant actions with explicit authority, confirmation, audit, and idempotency as appropriate.
- Keep core LAN operations independent of Internet connectivity.
- Automated tests, type checking, packaging, and physical/shop acceptance are distinct gates; passing one does not imply the others.

14. Retained workflow detail from the v2.0 baseline

The following non-conflicting details are retained from the source PDF. If wording in the old PDF differs from the canonical requirements above, the canonical requirements prevail; in particular, session controls are 30 minutes idle, 12 hours absolute, and a 10-minute critical reauthentication window.

Project scope, roles, and navigation

- Business scope includes mobile phone sales, repairs, parts and accessories; laptops; and computer repair, parts and accessories.
- The original navigation groups Dashboard, Sales/POS, Repair, Warranty, Stock, Purchases, Expenses, Cash/Day Start-End, Reports and Admin.
- Initial roles are Admin/Owner and Cashier/Staff; the original baseline anticipates future Manager/custom roles through the permission architecture.
- The original visual direction is a red-and-black theme with restrained blur/shadows and optional lightweight animation suitable for older laptops. Use separate task tabs/screens. Status must never be conveyed by color alone: pair green success, yellow warning, red danger/error, and blue/neutral information/history with text or other cues.
- Search covers item code, description, brand, model, barcode, category and IMEI/serial where applicable. Show in-stock items first, out-of-stock items below, and make negative stock visibly clear.

- Cashier capabilities include own completed-sale/reprint access, basic item entry/edit, permitted sale-time pricing and discounts, repair create/update/cancel, basic warranty claims, purchase entry without cost, stock/report view, day start/end/shift end, and pending expense/other-income entry. Admin controls costs, suppliers, stock adjustments, returns/exchanges, warranty decisions, approvals, users, permissions, reports, audit, backup/restore and settings.
- Repeated wrong-password attempts trigger temporary lockout. Cashier password reset requires Admin verification and a forced new password. Admin recovery uses a one-time offline Recovery Code and is audited. Critical Admin actions include restore, critical void/correction, permission changes, stock adjustment, damage/loss approval, purchase-cost correction and Day End reopen, subject to the canonical reauthentication policy.

Item and inventory details

- Sequential item codes may look like LVT0001; deleted codes are never reused and imported existing codes are preserved. Basic fields include Item Code, Description and Category 1; Category 2 and Item Types are dynamic/Admin-configurable.
- Multiple manufacturer/custom barcodes map to an item. Master sale price is optional. Historical sale lines retain the actual sale-time price.
- Phones/serialized items require unique IMEI/serial identity. Repair parts may be compatible across models. Display variants may include Original, High Quality, Phone Removed and Copy; displays use batch/quantity rather than serial tracking.
- Physical positive stock corrections use the latest appropriate purchase-lot cost. Stock movements also cover internal use and physical adjustments.

Sales, returns, purchases, repairs, and warranty details

- Sale flow: new sale -> search/scan -> cart -> quantity -> price -> discount -> total -> payment -> document type -> complete. Double-save protection and a visible processing state are required.
- Return charge is manually entered, has no fixed maximum, is separate Return Charge Income and is included in actual profit. Refund is calculated automatically; Admin may override it with audit.
  Cashier may mark exchange damage; Admin finalizes and may reverse damage through audited restock if resellable.
- An exchange links one Return and one entirely separate New Sale. The customer first receives the old-item refund, then pays the full new-sale amount. Example: old item 1000 with charge 200 gives an 800 refund; the separate new sale is 1500, so the customer pays 1500 and the net drawer movement is +700; return-charge income is 200.
- Cashier purchase entry may save basic invoice/quantities and increase stock immediately without a cost field. Admin privately completes pending line costs and may edit lines, quantities, date and reference. Purchase payment method is not asked or stored; supplier is optional/Admin-side. Completed costs become immutable purchase lots.
- Repair identifiers may use REP-000001 format. The source baseline lists statuses RECEIVED, CHECKING, WAITING FOR CUSTOMER, WAITING FOR PART and REPAIRING.
- Repair intake fields include Device Brand, Given Accessories, Special Note, Model, Customer Name, Contact No., Date In, Job No., Hand Over Date, Description/Amount, Repair Invoice, Advance Payment, Total, Reported Fault, Estimated Amount, Warranty Period and Authorized By.
- Repair estimates may differ from final charges; labour is separate and labour-only repair is supported. Removing an unused GOOD/resalable part restores its original FIFO lot; a damaged part enters Damage Review Pending. Cancellation returns unused/good parts and sends damaged parts to Damage/Loss. Historical payments require explicit reversal/refund, never deletion. Repair warranty is separate from product warranty.
- Product warranty types include NONE, calendar DAYS, calendar MONTHS, CHECKING ONLY and controlled CUSTOM rules. Manual/handwritten invoice details are captured in a normal sale by selecting Manual and recording its reference; separate Manual Warranty Registration is exception-only for legacy/external/late cases.

- Warranty claims are distinct from repairs. Admin decides Supplier Warranty Claim or Damage/Loss. Accepted supplier replacement is linked to the original claim/cost history; supplier rejection routes a shop-held item to Damage/Loss, while a customer-kept item gets no stock/profit write-off.
- Lost-bill recovery searches invoice/manual reference/IMEI/serial/customer/item/date. Reprint or duplicate never creates another sale or warranty.

Cash, reports, and financial details

- Opening cash corrections are Admin-only. Card/Online do not enter the physical drawer.
- The physical Cash Drawer excludes Card/Online. Cash In Hands counts approved shop inflows and outflows across payment methods; purchases/expenses do not automatically reduce Day-End physical drawer cash.
- Optional Other Cash In is a separate cash transaction that immediately updates Current Cash Balance and does not automatically link to main sales, stock, FIFO or main profit.
- Reload is entered once at Day End by brand (Dialog, Airtel, Mobitel or Hutch), cash-only, does not update live Current Cash Balance, is included in Day-End expected cash and is reported by brand. Any temporary estimated profit is not main profit.
- Day End shows expected cash, actual denomination total, difference, mandatory reason and optional note. Admin may reopen it. Missed/backdated sales retain original transaction date plus entry timestamp/user in audit; reports recalculate.
- Expense categories initially include Electricity, Rent, Internet, Transport, Repair Tools, Packing, Tea/Food, Other and Salaries; Admin may add categories. Admin corrections preserve history.
- Credit and repair settlements record date/time, amount, method, previous outstanding and remaining balance. Optional settlement proof/receipt can be generated and reprinted.
- Sales totals separate Cash, Card and Online plus combined total. Monthly reports include previous-month comparison charts. Target progress may exceed 100%; visual bars may stop at 100% while the percentage continues.
- Corrections retain old value, new value, user, time, reason and reference transaction. Sales profit is actual sale revenue less exact FIFO cost, independent of payment method. Profit combines Sales Profit, Repair Profit, Other Income and Return Charge Income less Expenses, Damage/Loss and other applicable losses. Cash In Hands, Profit and Cash Drawer remain distinct; credit/repair settlements are cash movements only.
- Audit coverage includes sales, edits, claims, discount limits, purchase corrections, stock adjustments, voids, returns/exchanges, Day End, approvals, permissions, settlements, payment corrections, supplier decisions, damage/restock reversals, backup/restore, manual bills/reprints and security events.

Backup continuity details

- Cashier sees Backup Status and Last Backup Time only; Admin sees verification details and restore points. An interrupted backup leaves the previous verified point valid; an interrupted restore must never be reported as successful.

Transaction and failure details

- Sale, Return, Exchange, Repair, Purchase and Payment are atomic: all effects commit or all roll back. Unsaved drafts cannot change stock, payment, FIFO or profit.
- Unique transaction IDs/idempotency protect crash and double-click retries. A power failure during commit leaves either the complete transaction or none. Concurrent edits use stale-record/version protection; Admin priority applies where specified.
- Duplicate IMEI/serial is blocked. Printer failure never rolls back a completed sale. Important financial, stock and warranty records are not hard-deleted.

Appendix A. Release and acceptance evidence gates

Use the release checklist as evidence, not as a claim that any gate has passed. Record PASS, FAIL, BLOCKED or NOT RUN with date, version, operator and evidence location. Automated tests do not replace shop acceptance.

Do not mark the release ready until applicable items have evidence recorded for the actual target machines and data. Automated tests do not replace shop acceptance. Mark each item PASS, FAIL, BLOCKED, or NOT RUN, with date, version, operator, and evidence location.

1. Build and software verification

- Current Python runtime matches backend metadata (>=3.13,<3.14); production dependencies are installed from the intended lock/constraints.
- pytest -q passes on the current source; record count, warnings, and environment.
- python -m compileall -q backend/app backend/alembic/versions passes.
- Alembic migration chain has one intended head; migration SQL/upgrade has been tested in staging; deployed DB revision is recorded.
- Focused frontend onboarding/API/discovery tests pass; no assertions are skipped to conceal failures.
- npx tsc -b passes.
- npm run build completes with the real app entry and normal tree-shaking/minification. Confirm Rollup resolves to the pinned 4.43.0 override; do not treat diagnostic output as a release bundle.
- Playwright smoke and required browser flows pass against a staging server.
- Tauri Windows build produces the intended signed/unsigned deliverables and launches; record artifact path, size, timestamp, and hash.
- No unresolved dependency, CSP, secret-scanning, or security review blockers remain.

2. Security and authorization

- Cashier API responses and exports never reveal purchase/supplier cost, FIFO unit cost, profit, internal pricing rationale, or detailed backup/restore records.
- Backend rejects unauthorized actions regardless of UI visibility; verify permission removal takes effect.
- Session idle timeout is 30 minutes, absolute lifetime is 12 hours, and critical reauthentication behavior matches the required 10-minute policy.
- Admin recovery, password reset, forced password change, token revocation, and branch isolation work as specified.
- Test security-sensitive actions: critical reauthentication, stock/purchase corrections, permission changes, day reopen, backup restore, overrides, and export.
- Append-only audit/history survives attempted update/delete at database level for protected tables.
- Secrets are unique per environment, protected from cashier access, absent from bundles/logs/docs, and recoverable where backup decryption requires them.
- Confirm Windows account privileges, firewall scope, PostgreSQL exposure, disk encryption status, Tauri CSP, and update/patch process.

3. LAN, role, and startup

- Cashier laptop hosts PostgreSQL and FastAPI; owner/client laptop may be powered off without stopping POS operation.
- Clients never connect directly to PostgreSQL; server firewall allows only the required private-LAN API traffic.
- Fresh client checks only 127.0.0.1 to determine whether the local machine is the server.
- Saved machine role is preserved on later launch; client discovers the current server without a hardcoded IP.
- Server identity and setup state are checked on discovery and manual address entry; an unverified/mismatched server cannot enter READY.
- Bootstrap persists the session and refreshes verified discovery before READY. Failed verification is retryable without resubmitting bootstrap.

- Server starts after reboot only when PostgreSQL is ready; test service recovery and client reconnect.
- Test DHCP/address change, DNS/mDNS, firewall, LAN interruption, concurrent client access, and replacement-server identity/recovery.
- Core workflows continue with Internet disconnected.

4. Sales, inventory, and finance

- Cash, Card, Online, split payments, typed prices, discount authority/approval, and Admin-only credit sale are tested.
- Invoice numbers are atomic and never reused; duplicate-click and timeout retry create one transaction.
- Printer failure after commit does not roll back sale; reprint records a separate document event.
- FIFO is exact under multiple lots, negative stock, subsequent purchases, corrections, returns, repairs, and concurrency.
- Negative-stock reconciliation is deterministic and later purchases retain full lot quantity/cost.
- Purchase cost/date/quantity/reference corrections are authorized, version-checked, audited, and replay FIFO safely.
- Partial returns, seven-day window/Admin override, GOOD lineage restoration, damaged loss, refunds, return charges, and exchange accounting reconcile.
- Purchase entry keeps cost private from Cashier; pending Admin cost completion creates correct lots.
- Repair advance/final/split payments, FIFO parts, GOOD part returns, damage review, delivery override, cancellation, and immutable payment history work.
- Warranty dates/statuses, all supported rule types, claims, rejection, supplier replacement, damage outcome, exception registration, lost-bill recovery, and sale-time snapshots work.
- Day Start denominations, Shift End vs Day End, Continue Today/auto-close, Admin reopen, one reload per day, Day-End differences, expenses/other-income approval, Cash Drawer, and Cash In Hands reconcile.
- Customer history, credit/repair settlements, proof, reprints, reports, target snapshots, and PDF/Excel exports are role-appropriate and correct.

5. Backup, restore, and data migration

- Backup lifecycle is recorded; only VERIFIED backups are restore candidates.
- Backup encryption/integrity verification succeeds; recovery key custody and offline recovery are proven.
- Emergency local backup is physically/logically separate from database working files.
- Interrupt upload/connectivity, retry queued backups, and prove ordinary LAN sales do not depend on Google Drive.
- Restore flow enforces Admin reauthentication, safety backup, explicit confirmation, selected verified point, post-restore integrity, and immutable audit event.
- Perform restore on an isolated replacement server and prevent simultaneous writers/split brain.
- Complete historical spreadsheet import with Admin preview/approval, provenance, duplicate handling, and financial/stock reconciliation before go-live.

6. Windows hardware and operations

- Verify actual Rust stable/Cargo, Tauri CLI, Microsoft C++ Build Tools, and WebView2 prerequisites.
- Install and launch the actual Tauri package on target Windows machines; verify app restart, update, and uninstall behavior.
- USB barcode scanner input is correct for supported item/serial flows.
- ESC/POS receipt and A5 warranty printers work; unplug/reconnect does not corrupt transactions.
- Reboot after installation and verify PostgreSQL/API startup, readiness, logs, and backup schedules.
- Document operator procedures, escalation, recovery contact, data retention, and supported upgrade path.

7. Go-live authorization

- Use synthetic data in staging; complete the acceptance matrix before production install.
- Run an authorized parallel period with the prior process remaining source of truth.
- Verify a recent encrypted backup and successful restore drill before cutover.
- Obtain Admin/owner sign-off for data reconciliation, permissions, security, operations, and recovery.
- Archive the tested build identity, source revision, dependency versions, migration revision, configuration template, test logs, artifact hashes, and release checklist.
- Confirm the project is not described as production-ready while any mandatory item is FAIL, BLOCKED, or NOT RUN.

Repair Status and Customer-Document Clarifications

Complete repair status lifecycle

Supported repair statuses (the complete status set; transitions follow the authorized workflow): RECEIVED, CHECKING, WAITING FOR CUSTOMER, WAITING FOR PART, REPAIRING, READY, DELIVERED, RETURNED TO CUSTOMER, and CANCELLED.

Preserve repair status history. Do not remove or rename any of these statuses.

Customer-facing document privacy

Customer-facing invoices, receipts, and other customer-facing documents show the final selling price only; never show buy cost, FIFO cost, profit, or internal pricing rationale.

Deployment, Server Discovery & Disaster Recovery

Installer, machine roles, and topology

The same supported Lovely Tech POS installer/executable must offer SERVER and CLIENT installation. Machine role (SERVER / CLIENT) is independent of signed-in user role (ADMIN / CASHIER). The Cashier laptop hosts PostgreSQL and FastAPI as the always-on shop server. Owner/Admin laptops are clients and may be switched off without stopping the server or other clients.

Client and server communicate through the private-LAN REST API. Clients never connect directly to PostgreSQL or receive database credentials. Normal LAN transactions do not depend on Internet/cloud availability while the server, its database, and LAN remain operational.

Shop, server, and machine identity

Shop Identity identifies the logical shop and its business history and remains stable across router changes, device replacement, and verified restore. Server Identity identifies the persistent authorized server for that shop and one active writer. Machine Identity identifies an installation for local configuration; it is not Shop Identity, user identity, or write authority. Replacing a client creates a new Machine Identity while reconnecting to the same shop.

Clients bind to verified logical identity, not permanently to an IP address. A restored database must be reconciled with the existing Shop Identity and authorized Server Identity before advertising service or accepting writes. Do not silently create another shop, clone an unverified identity, or merge independently writable databases.

Zero-IP discovery and trust

Clients automatically discover the genuine Lovely Tech POS service on the LAN and reconnect after router replacement or DHCP/IP changes without reinstall or manual IP edits. Do not hardcode LAN IP addresses.

Verify product, supported protocol/API version, setup state, and persistent identity using the service advertisement and server discovery endpoint. An open port, including 8000, is not proof of a genuine server. Reject incomplete, mismatched, unsupported, or ambiguous identities. Manual address entry, if available, is an Admin-controlled fallback subject to the same checks; it is not a READY bypass.

Multiple servers claiming one active Shop Identity, conflicting identity, or split-brain evidence must fail closed for writes. Keep candidates isolated until an authorized Admin resolves single-writer ownership and records the resolution.

Guided Setup and Recovery Scenarios

Installer/setup wizard

On installation, detect a verified existing shop where possible and offer authorized paths to connect to an existing shop, create a new shop, or restore an existing shop.

New-shop setup confirms that a new shop is intended, guides creation of the first Admin account, and requires security decisions including unique credentials/secrets and recovery-material custody. Recovery, server identity reassignment, and privileged configuration are Admin-only and require critical reauthentication and audit. Cashiers receive no recovery controls.

Ordinary operators must not need to enter IP addresses, PostgreSQL credentials, Python/Node settings, database URLs/connection strings, or manually run migrations. Do not weaken privilege checks or expose secrets to achieve zero setup. On failed verification, stop before writes and explain the recovery state safely.

Mandatory disaster-recovery acceptance scenarios

- Router replacement and server DHCP/IP change: clients rediscover and reconnect without reinstall or manual address edits.
- Server laptop failure: install the same application on a replacement server, restore the latest VERIFIED backup, reconcile Shop/Server Identity, verify health/integrity, allow one writer, then reconnect clients.
- Client laptop replacement: install as CLIENT with a new Machine Identity and reconnect to the same verified shop without modifying business history.
- Server and all clients replaced: restore one authorized server, install new clients, and verify they all resolve the same logical shop.
- New shop with no database: create a new shop and first Admin; do not present bootstrap as restore.
- Owner/Admin laptop off: the cashier server and remaining clients continue normal LAN operation.
- No VERIFIED backup, failed restore, missing recovery key, identity mismatch, duplicate server, or split brain: do not report success or enable unverified writes; preserve available data/safety copies and give an explicit next step.

These are acceptance requirements, not a statement that end-to-end scenarios have passed.

Backup, Offline Operation, and Safe Maintenance

Transaction persistence versus backup transport

A sale committed to the live server database is transaction persistence. An encrypted backup awaiting cloud upload is a separate recovery copy, not a transaction queue or evidence that a committed sale was lost.

When server/database and LAN remain healthy, Internet or Google Drive loss must not discard or roll back committed sales or unnecessarily block normal sales. Backup-at-close policy is separate and may require an available verified local/USB/external/network fallback.

Keep local backups encrypted and integrity-verified. Durably queue failed cloud uploads, retry when connectivity returns, and mark remote sync complete only after upload and verification. Preserve the local verified copy and show pending/failed sync state.

Restore controls and recovery limits

Only VERIFIED backups may be restored. Require Admin permission and critical reauthentication, a verified safety backup of the target, explicit confirmation, a selected verified point, post-restore integrity and identity checks, and immutable audit/history.

Recovery is limited to retained verified copies and available decryption material. Do not promise recovery if all usable verified backups or their required decryption secret are lost.

Upgrade and compatibility

Use an integrity-checked supported upgrade path that preserves Shop Identity and business data. Check client/server protocol compatibility and block incompatible versions with an actionable update path.

Before upgrade, verify a recent backup and recovery material. Test package and migrations on staging; apply supported migrations once through the server workflow; then check health/readiness, identity, and audit. Preserve one active writer. Do not promise downgrade safety where migrations are irreversible.

Implementation Evidence and Remaining Gaps

Implementation status — not acceptance evidence

Reviewed source contains separate Windows server install/start scripts and a separate Tauri client packaging script; one role-aware installer and complete connect/new/restore wizard are not demonstrated.

The frontend stores machine role and a CLIENT identifier locally. The backend has a durable instance_id file and branch_code; mDNS advertises identity fields and the client compares them with /discovery/info. Discovery currently chooses the first verified candidate; duplicate-server rejection and exclusive active-writer enforcement are not demonstrated.

The instance_id file is outside the PostgreSQL dump, and branch_code is the current shop/branch identifier; a separate stable Shop Identity and safe restore rebind are not demonstrated. Database restore alone therefore does not prove identity recovery.

Backup source creates and locally verifies encrypted dumps; cloud failure may be recorded PENDING_SYNC and a manual sync endpoint exists. Durable automatic retry and remote-object verification are not demonstrated. Router/DHCP changes, replacement-host/all-device restore, cloud retry, split-brain prevention, and upgrade acceptance remain unverified.